

TU DORTMUND UNIVERSITY

Amortized Bayesian Inference for Double Responses in Speeded Decision Tasks

Simulation Based Inference with a Lateral Inhibition Racing Diffusion
Model

MOSFIQUN NAHID HASSAN	235632
MD. JUBIRUL ALAM	230980
SHOWAIBUZZAMAN SHOWAIBUZZAMAN	258947

Course: Simulation Based Inference

Final report | August 2026

Contents

1	Introduction	2
1.1	Motivation	2
1.2	Problem	2
1.3	Modelling idea	2
2	Data	3
2.1	Data source and experimental design	3
2.2	Preprocessing and coding	3
3	Statistical model	4
3.1	Lateral-inhibition racing diffusion model	4
3.2	Priors	5
3.3	Prior predictive check	5
4	Approximator	6
4.1	BayesFlow architecture	6
4.2	Why the data representation matters	6
5	Training	6
5.1	Training regime	6
5.2	Convergence and checkpoint selection	7
6	Diagnostics	8
6.1	Parameter recovery	8
6.2	Simulation-based calibration	9
6.3	Posterior contraction and z-score diagnostics	10
7	Inference on empirical data	11
7.1	Posterior parameter estimates	11
7.2	Posterior predictive checks	12
7.3	Double-response-time predictions	15
8	Discussion and limitations	18
9	Conclusion	18
10	Reflection on Our Learning	19

1 Introduction

1.1 Motivation

Speeded decision tasks are usually summarized by two observations: the first choice and its response time. These measurements are useful, but they do not always reveal what happened immediately after the first decision. A participant can sometimes make a second response shortly after the first one. Evans et al. [1] call this behaviour *double responding*.

Double responses are often removed during preprocessing, even though they may contain information about the decision process. In an evidence accumulation model, the competing alternatives can continue to accumulate evidence after one alternative has reached the response threshold. A rapid second threshold crossing can therefore provide an additional constraint on the model. The aim of this project is to use this additional information rather than discard it.

1.2 Problem

The main problem is to infer the latent parameters of a decision model from data containing choices, response times, and double responses. We use the data from Experiment 1 of Evans et al. [1], where four participants each contributed approximately 10,000 trials. Two participants performed under speed emphasis and two under accuracy emphasis.

The model should reproduce not only accuracy and response times, but also how often double responses occur and whether they are more common after correct or erroneous first responses.

1.3 Modelling idea

We use a two-alternative racing diffusion model with lateral inhibition. Each alternative accumulates noisy evidence over time. The first accumulator to reach the threshold produces the first response. Importantly, accumulation does not stop immediately after that response. During the next 250 ms, the competing accumulator may reach the threshold as well, producing a double response.

The five parameters inferred in this project are the drift rates v_A and v_B , lateral-inhibition strength β , threshold b , and non-decision time t_0 . We build a simulator for this process and train a BayesFlow neural posterior estimator on simulated datasets. The trained estimator is then applied to the empirical participant data.

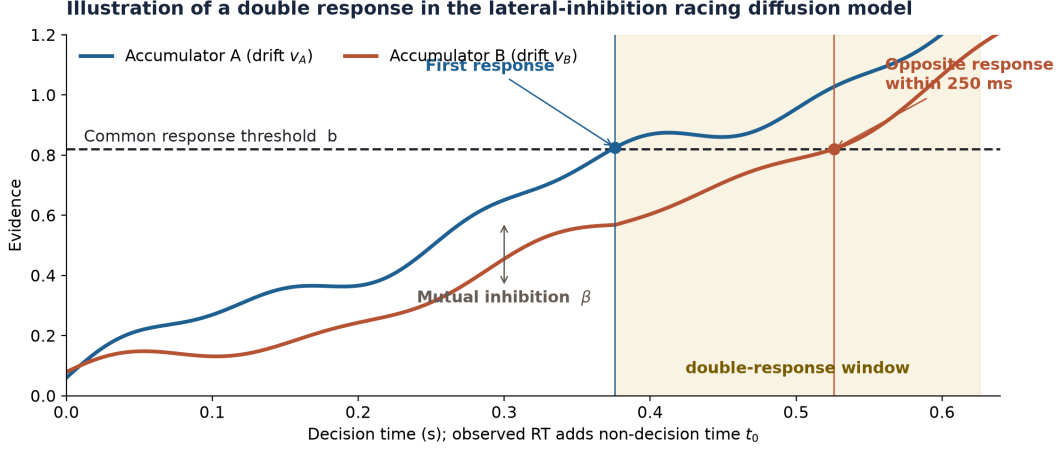


Figure 1: Illustration of the decision process used in the project. Two competing evidence trajectories accumulate toward a response threshold. The first threshold crossing produces the initial response; a second crossing by the competing accumulator within the 250 ms double-response window produces a double response.

2 Data

2.1 Data source and experimental design

The empirical data come from Experiment 1 described by Evans et al. [1]. The experiment was based on the data of Dutilh et al. [2]. Each participant completed 25 blocks of trials, with 10,000 trials per participant in the dataset used by Evans et al. The original study was not designed around double responding; the phenomenon was identified retrospectively in the recorded responses. This makes the dataset useful for testing whether double responses provide information beyond standard choice and response-time measurements.

Participants 1 and 2 were in the speed-emphasis condition, while participants 3 and 4 were in the accuracy-emphasis condition. This distinction is reflected in the empirical response patterns: the speed participants were faster and less accurate, and they also showed more double responses.

2.2 Preprocessing and coding

The empirical files contain the condition, correctness of the first response, first-response time, double-response indicator, and second-response time. We applied the same preprocessing procedure to the simulated data, removing trials with first-response times below 250 ms.

The retained sample sizes were 9,874, 9,963, 9,998, and 9,998 trials for participants 1–4, respectively. The first response was represented numerically as 0 for a correct response and 1 for an error. For trials without a second response, the missing second-response time was replaced by zero for numerical processing. The separate double-response indicator tells the network whether this zero is a placeholder or whether a second response was actually observed.

Table 1: Descriptive statistics after the 250 ms response-time exclusion.

Participant	Condition	Retained n	Accuracy	Mean RT (s)	Double response
P1	Speed	9,874	0.7376	0.4384	0.0191
P2	Speed	9,963	0.8266	0.4948	0.0256
P3	Accuracy	9,998	0.9482	0.5679	0.0003
P4	Accuracy	9,998	0.9288	0.5297	0.0007

Double responses were rare overall (1.91%, 2.56%, 0.03%, and 0.07% of retained trials for P1–P4), but they were far more common after errors (6.25%, 12.91%, 0.58%, and 0.98%, respectively). For P1, 189 double responses were observed; for P2, 255; for P3, only 3; and for P4, 7. This imbalance is important when interpreting the double-response-time distributions for P3 and P4.

3 Statistical model

3.1 Lateral-inhibition racing diffusion model

For each trial, two accumulators represent the competing responses: accumulator A for the correct response and accumulator B for the error response, matching the 0/1 correctness coding introduced in Section 2. Let $X_A(t)$ and $X_B(t)$ denote their accumulated evidence. The dynamics are simulated with a discrete time step of $dt = 0.002$ s. The drift of each accumulator is affected by evidence accumulated by the competing alternative.

Conceptually, the lateral-inhibition mechanism can be written as

$$dX_A(t) = [v_A - \beta X_B(t)] dt + \sigma dW_A(t),$$

with the analogous equation for accumulator B . Here v_A and v_B control the mean evidence accumulation, β controls competition between accumulators, and the diffusion coefficient was fixed at $\sigma = 1$ to identify the scale. The implementation uses the project simulator and the same parameterization for simulation and inference.

A first response occurs when either accumulator reaches the threshold b . The response time additionally contains the non-decision component t_0 . After the first response, both accumulators continue to evolve for a maximum of 250 ms. If the competing accumulator reaches b during this interval, the trial is labelled as a double response and the delay to the second response is recorded.

3.2 Priors

We used explicit proper priors for all five parameters. The priors were truncated normal distributions:

$$\begin{aligned} v_A &\sim \mathcal{N}(3.0, 1.5^2), & v_A > 0.05, \\ v_B &\sim \mathcal{N}(1.5, 1.0^2), & v_B > 0.05, \\ \beta &\sim \mathcal{N}(3.0, 1.5^2), & \beta > 0, \\ b &\sim \mathcal{N}(1.4, 0.4^2), & b > 0.3, \\ t_0 &\sim \mathcal{N}(0.3, 0.15^2), & 0.05 < t_0 < 0.6. \end{aligned}$$

The number of trials in training simulations was varied rather than fixed to a single value. The training distribution used 500, 1,000, 2,500, 5,000, and 10,000 trials with probabilities 0.30, 0.25, 0.20, 0.15, and 0.10, respectively. Thus, the network saw datasets covering the empirical sample-size range while also seeing the exact 10,000-trial setting.

3.3 Prior predictive check

Before training, we simulated 200 datasets from the prior predictive distribution. The median simulated accuracy was 0.7797, the mean response time was 0.6676 s, and the median double-response probability was 0.0057, with corresponding 5–95% ranges of [0.3153, 0.9780], [0.4293, 1.0909] s, and [0, 0.2770]. Every empirical participant value fell inside its respective range, and the observed double-response rates occupied approximately the 29th–65th percentiles of the prior-predictive distribution, confirming that the priors covered the empirical regime without narrowly targeting it.

The prior predictive check was performed for all three central observables: accuracy, response time, and double-response rate.

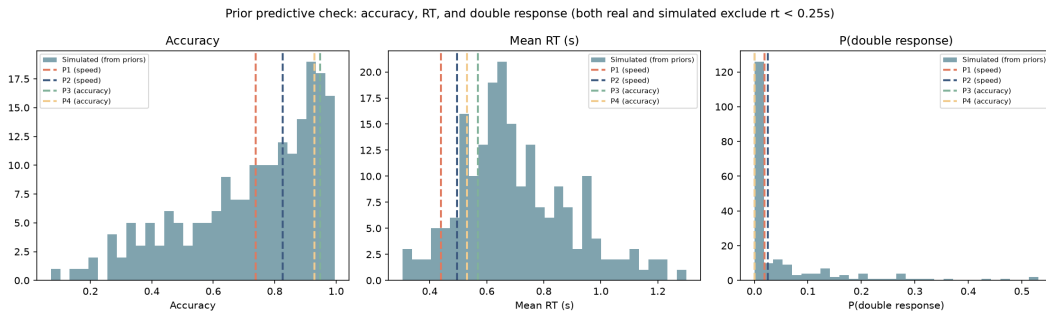


Figure 2: Prior-predictive distributions from the 200 simulated datasets for accuracy, mean response time, and double-response probability, with the four empirical participants marked for comparison.

4 Approximator

4.1 BayesFlow architecture

The simulator was connected to a BayesFlow [3] neural posterior estimator. The observation adapter maps the simulated variables to the representation used by the inference network. The summary network compresses the variable-length trial-level data into a fixed-dimensional representation, and the inference network maps this representation to an approximate posterior over the five model parameters.

The coupling-flow posterior network uses six dual-coupling blocks with random permutations between them, following the BayesFlow coupling-subnet defaults from the installed BayesFlow 2.0.12 implementation. The coupling MLP uses widths (128, 128), hard-SiLU activation, orthogonal kernel initialization, no residual connection, dropout 0.05, and no normalization. The spline transform uses 16 bins with default domain $[-3, 3]$ for both input and output dimensions, minimum bin width and height 0.1, minimum overall width and height 1.0, and rational-quadratic splines.

The summary network is a Set Transformer designed to handle the trial-level structure. It uses two self-attention blocks with 64-dimensional embeddings and 4 attention heads each, 32 inducing points to keep attention cost manageable for datasets as large as 10,000 trials, and 4 pooling seeds in the final aggregation step, producing a 10-dimensional summary vector. The attention blocks use the corresponding BayesFlow implementation with SwiGLU feed-forward transformations and pre-layer-normalized RMS normalization.

4.2 Why the data representation matters

The network receives both the first-response information and the double-response information. In particular, the second-response time is missing when no second response occurs. Replacing a missing value by zero alone would be ambiguous, so the separate double-response indicator is retained. This allows the network to distinguish “no second response” from an observed second response whose numerical delay is close to zero.

5 Training

5.1 Training regime

Training was performed online. In each training step, new parameters were sampled from the priors and new datasets were generated by the simulator. Therefore, a conventional validation loss based on a fixed held-out validation dataset was not used. Instead, we assessed convergence using the training loss together with simulation-based calibration, parameter recovery, and posterior-contraction diagnostics on independently simulated validation datasets.

The initial training phase used 40 epochs with a batch size of 4 and 400 batches per epoch. Because the loss was still decreasing, training was continued for two additional 20-epoch phases. The first continuation used a learning-rate target of 10^{-4} and the second used

5×10^{-5} . Both continuation phases used 4 datasets per batch and 400 batches per epoch. The effective schedule extracted from the saved checkpoints used a short warm-up followed by cosine decay. For the epoch-40 to epoch-60 continuation, the schedule started at 10^{-5} , warmed up to 10^{-4} over 400 steps, and then decayed over the remaining training steps. For the epoch-60 to epoch-80 continuation, the corresponding values were 5×10^{-6} and 5×10^{-5} .

Table 2: Training and continuation budget.

Stage	Epochs	Batch size	Batches/epoch	LR target
Initial training	40	4	400	original schedule
Continuation 1	20	4	400	1×10^{-4}
Continuation 2	20	4	400	5×10^{-5}

5.2 Convergence and checkpoint selection

The epoch-40 model was continued to epoch 60 because its loss had not stabilized. During the 40–60 continuation, the final loss was 0.4586. The mean loss in the preceding 20% of the continuation was 0.8041, compared with 0.6349 in the final 20%, showing a substantial downward trend.

A further continuation to epoch 80 reduced the final loss to 0.1329. The mean loss decreased from 0.3320 in the preceding 20% to 0.2746 in the final 20%. Thus, lower training loss alone did not provide a reason to stop at epoch 80.

However, the final model used for empirical inference was epoch 60 rather than epoch 80. The reason was calibration rather than the training loss. The epoch-80 model achieved a lower loss, but its SBC and recovery diagnostics did not improve consistently; in particular, calibration for β became worse and recovery for some parameters remained weaker. We therefore selected epoch 60 as the better compromise between fit and calibrated inference.



Figure 3: Training loss during the initial 40-epoch phase.



Figure 4: Training loss during the epoch-40 to epoch-60 continuation.

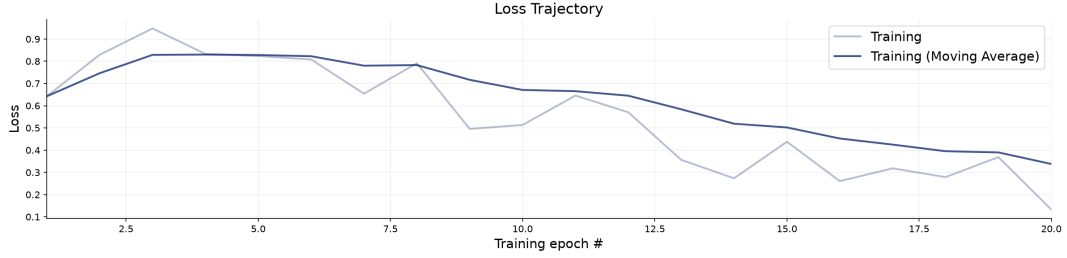


Figure 5: Training loss during the epoch-60 to epoch-80 continuation. The loss is lower than at epoch 60, but the additional reduction did not translate into uniformly better calibration.

6 Diagnostics

6.1 Parameter recovery

We evaluated the trained estimator on 200 held-out simulated datasets with 10,000 trials each and drew 500 posterior samples per dataset. Parameter-recovery plots compare posterior estimates with the true parameters used to generate each dataset.

For the epoch-40 model, the approximate recovery correlations for $(v_A, v_B, \beta, b, t_0)$ were 0.890, 0.696, 0.972, 0.846, and 0.970, respectively. At epoch 60 – the checkpoint used for empirical inference – they were 0.888, 0.646, 0.976, 0.858, and 0.974. At epoch 80 they were 0.891, 0.663, 0.982, 0.857, and 0.978. These results show strong recovery for most parameters, while v_B remains the more difficult parameter to recover.

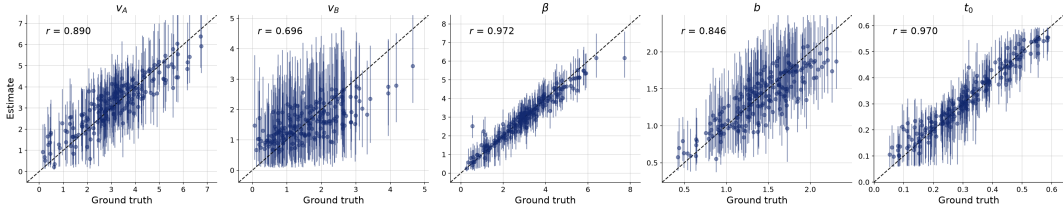


Figure 6: Parameter recovery for the epoch-40 model. Points compare posterior estimates with the true simulated parameters.

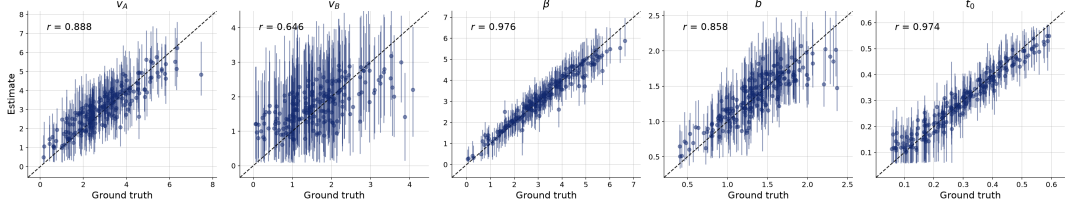


Figure 7: Parameter recovery for the epoch-60 model, the checkpoint used for all empirical inference.

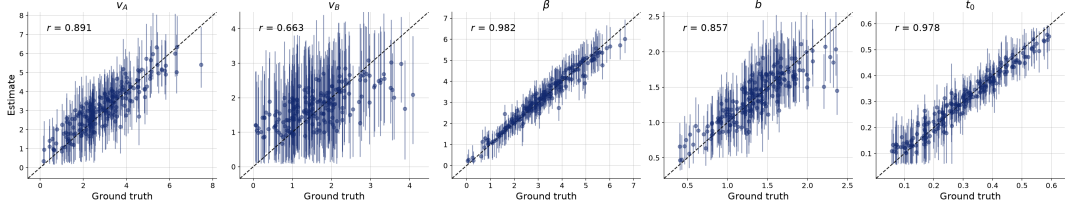


Figure 8: Parameter recovery after the additional training to epoch 80.

6.2 Simulation-based calibration

Simulation-based calibration (SBC) checks whether the posterior ranks of the true parameter values are consistent with a calibrated posterior. We used ECDF plots based on independently simulated validation datasets.

The SBC results were used together with recovery rather than interpreted in isolation. A model can show high parameter-recovery correlations while still producing posteriors that are too narrow or systematically biased. The epoch-80 model did not provide a uniform improvement in this respect, which contributed to our choice of epoch 60 for real-data inference.

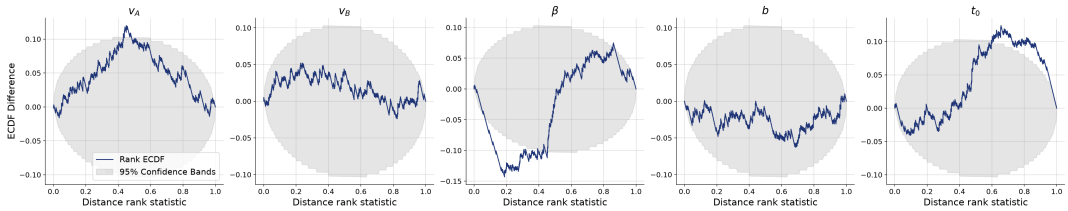


Figure 9: SBC ECDF diagnostics for the epoch-40 model, before either continuation phase.

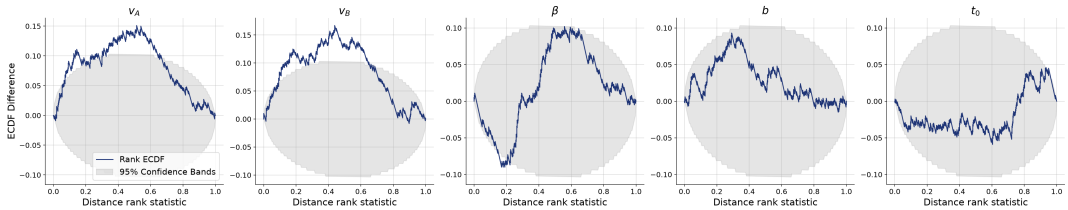


Figure 10: SBC ECDF diagnostics for the epoch-60 model. Deviations from the expected uniform-rank behaviour indicate potential calibration problems.

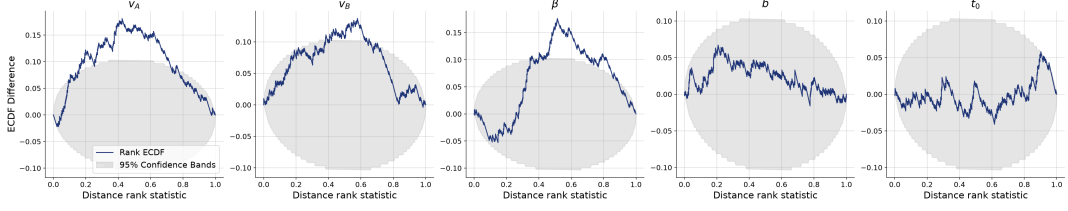


Figure 11: SBC ECDF diagnostics for the epoch-80 model. The comparison with epoch 60 was used when selecting the final checkpoint.

6.3 Posterior contraction and z-score diagnostics

Posterior contraction was assessed by comparing posterior uncertainty with the uncertainty in the prior, while the z-score diagnostic examines whether standardised posterior errors behave as expected. These diagnostics complement recovery by asking whether the estimator becomes appropriately more certain when the data contain information about a parameter.

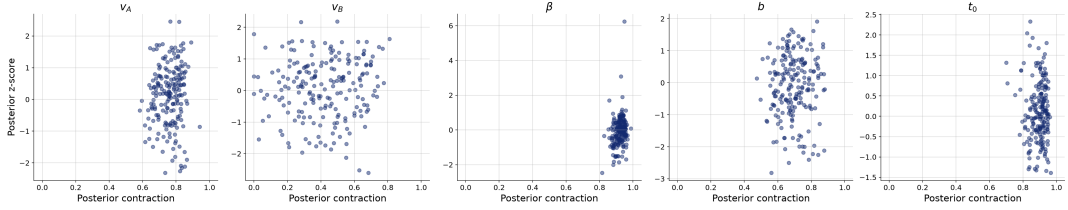


Figure 12: Posterior contraction and z-score diagnostics for the epoch-40 model.

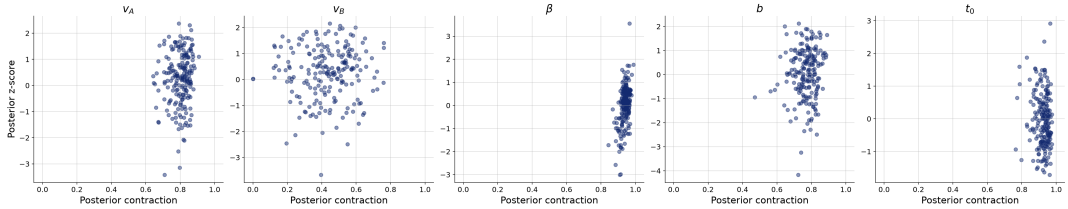


Figure 13: Posterior contraction and z-score diagnostics for the epoch-60 model.

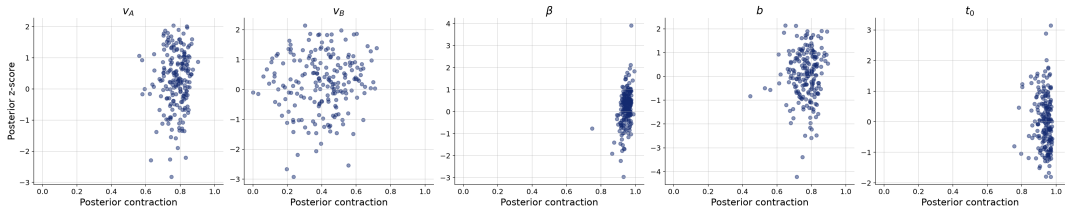


Figure 14: Posterior contraction and z-score diagnostics for the epoch-80 model. Higher contraction did not consistently correspond to better-centered z-scores.

Taken together, the diagnostics reveal different levels of inferential quality across parameters. v_B shows the weakest parameter recovery, whereas β has a very high recovery correlation but exhibits less satisfactory calibration in the SBC diagnostics. Although training to epoch 80

improves some recovery correlations, the calibration of β becomes worse. The contraction–z-score plots further show that stronger posterior contraction does not necessarily imply better-centered posterior estimates. Considering these diagnostics jointly, the epoch-60 checkpoint provides the better overall compromise between recovery, calibration, and posterior concentration.

7 Inference on empirical data

7.1 Posterior parameter estimates

The epoch-60 approximator was applied separately to the four empirical participants. The posterior means and 95% credible intervals are shown below.

Table 3: Posterior parameter estimates for the four participants.

Participant	Parameter	Mean	2.5%	97.5%
P1	v_A	4.179	2.809	5.372
	v_B	2.249	0.776	3.731
	β	6.662	5.476	7.971
	b	0.815	0.522	1.081
	t_0	0.248	0.188	0.300
P2	v_A	4.847	3.678	6.201
	v_B	1.971	0.607	3.329
	β	6.391	5.360	7.644
	b	0.783	0.529	1.008
	t_0	0.329	0.287	0.381
P3	v_A	4.964	3.861	6.304
	v_B	1.217	0.165	2.554
	β	4.907	4.251	5.713
	b	1.407	1.128	1.759
	t_0	0.279	0.239	0.321
P4	v_A	4.736	3.622	6.102
	v_B	1.287	0.182	2.724
	β	4.794	4.104	5.569
	b	1.384	1.089	1.743
	t_0	0.231	0.173	0.279

The estimates show a clear qualitative difference between the speed and accuracy participants. The speed participants have substantially smaller thresholds than the accuracy participants, and their lateral-inhibition parameters are also higher. These posterior estimates should not be interpreted independently: drift rates, threshold, non-decision time, and inhibition jointly determine the observed choice and response-time behaviour, and some parameters – especially v_B – remain substantially uncertain.

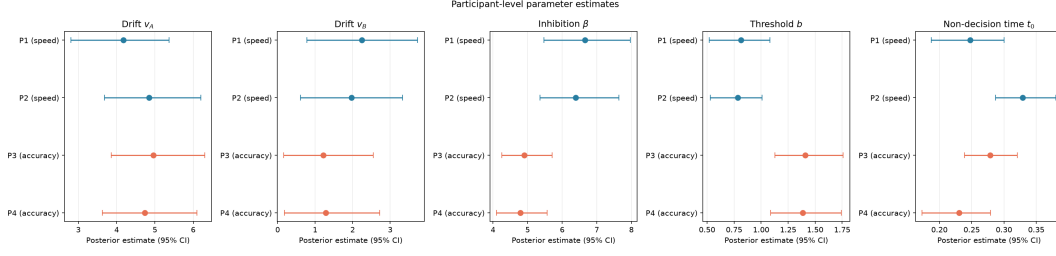


Figure 15: Posterior parameter estimates for the four participants, shown with 95% credible intervals.

7.2 Posterior predictive checks

Posterior predictive checks (PPCs) compare the empirical data with datasets generated from posterior parameter draws. We used 100 posterior parameter draws for the predictive simulations. The checks considered accuracy, mean response time, overall double-response rate, conditional double-response rates, response-time distributions for correct and incorrect responses, and double-response times.

Table 4: Posterior predictive checks for accuracy and mean response time. Intervals are 90% posterior predictive intervals.

Participant	Accuracy			Mean RT (s)		
	Real	Pred.	90% PI	Real	Pred.	90% PI
P1	.7376	.7353	[.6700,.7903]	.4384	.4315	[.3891,.4740]
P2	.8266	.8228	[.7767,.8753]	.4948	.4900	[.4635,.5224]
P3	.9482	.9479	[.9172,.9769]	.5679	.5614	[.5243,.6007]
P4	.9288	.9333	[.8981,.9632]	.5297	.5254	[.4879,.5630]

Accuracy and mean response time are reproduced closely for all four participants. The empirical values fall within the predictive intervals in every case, indicating that the posterior distributions contain parameter values capable of generating the main choice and response-time summaries.

Table 5: Posterior predictive checks for double responding.

Participant	Overall		Given correct		Given error	
	Real	Pred.	Real	Pred.	Real	Pred.
P1	.0191	.0408	.0037	.0170	.0625	.1098
P2	.0256	.0500	.0039	.0156	.1291	.2130
P3	.0003	.0007	.0000	.0000	.0058	.0150
P4	.0007	.0012	.0000	.0001	.0098	.0175

The model captures the important qualitative pattern: double responses are much more common after errors than after correct responses. However, the posterior predictive mean is systematically higher than the observed overall double-response rate, especially for P1 and P2.

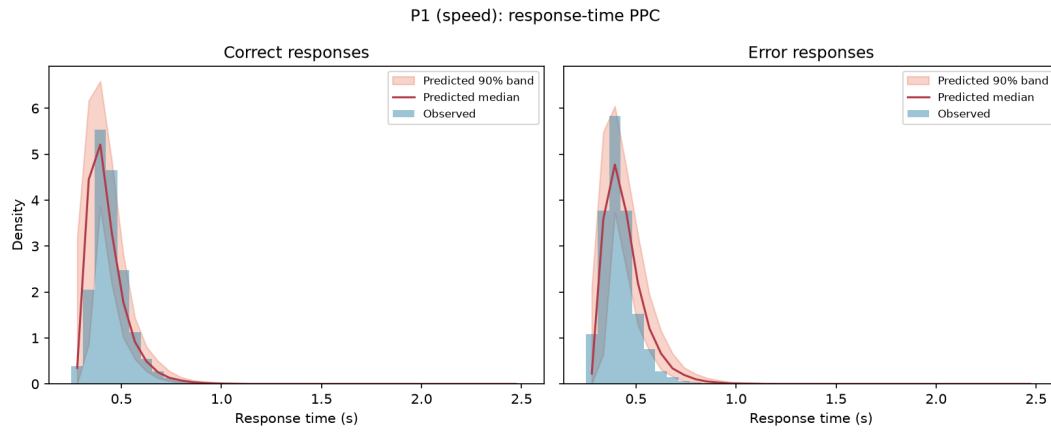


Figure 16: Posterior predictive response-time distribution for participant 1, separated by correct and incorrect first responses. The predictive spread represents uncertainty across posterior draws.

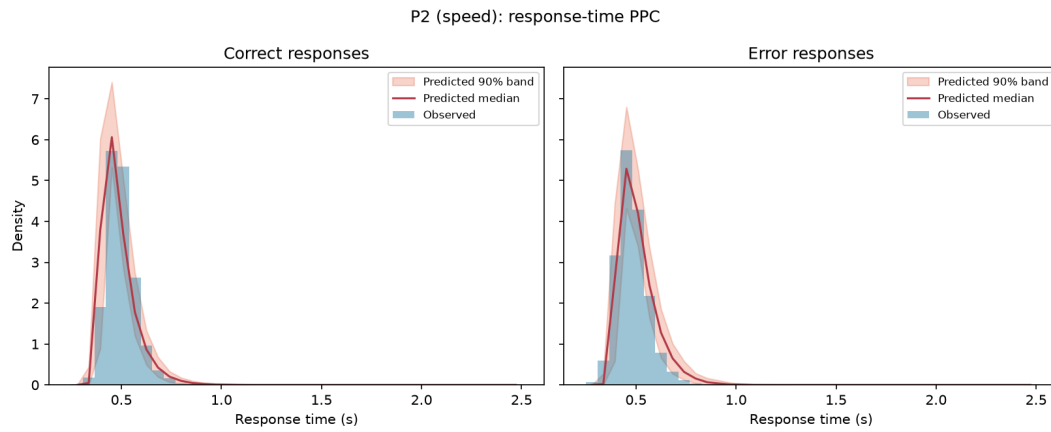


Figure 17: Posterior predictive response-time distribution for participant 2, separated by correct and incorrect first responses.

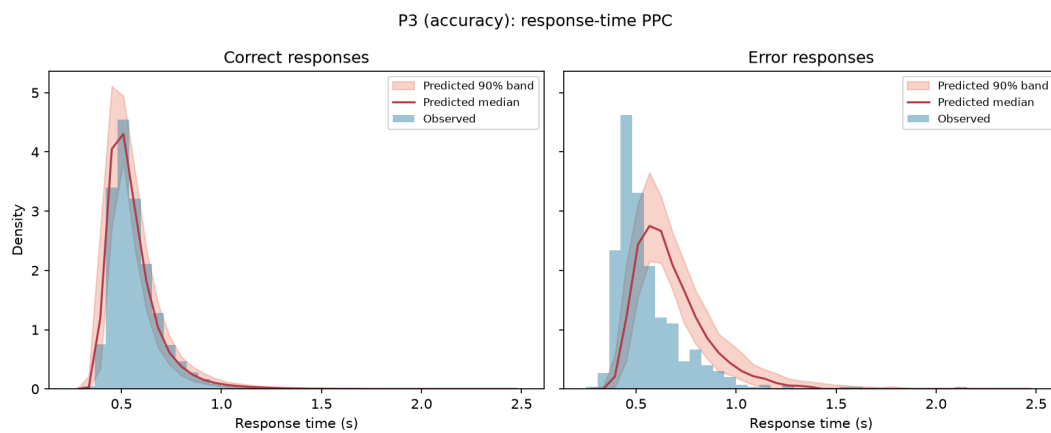


Figure 18: Posterior predictive response-time distribution for participant 3, separated by correct and incorrect first responses.

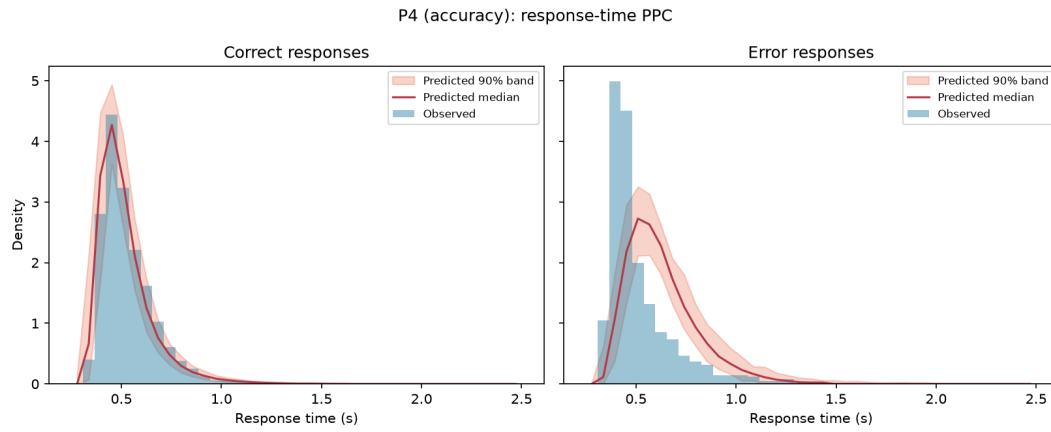


Figure 19: Posterior predictive response-time distribution for participant 4, separated by correct and incorrect first responses.

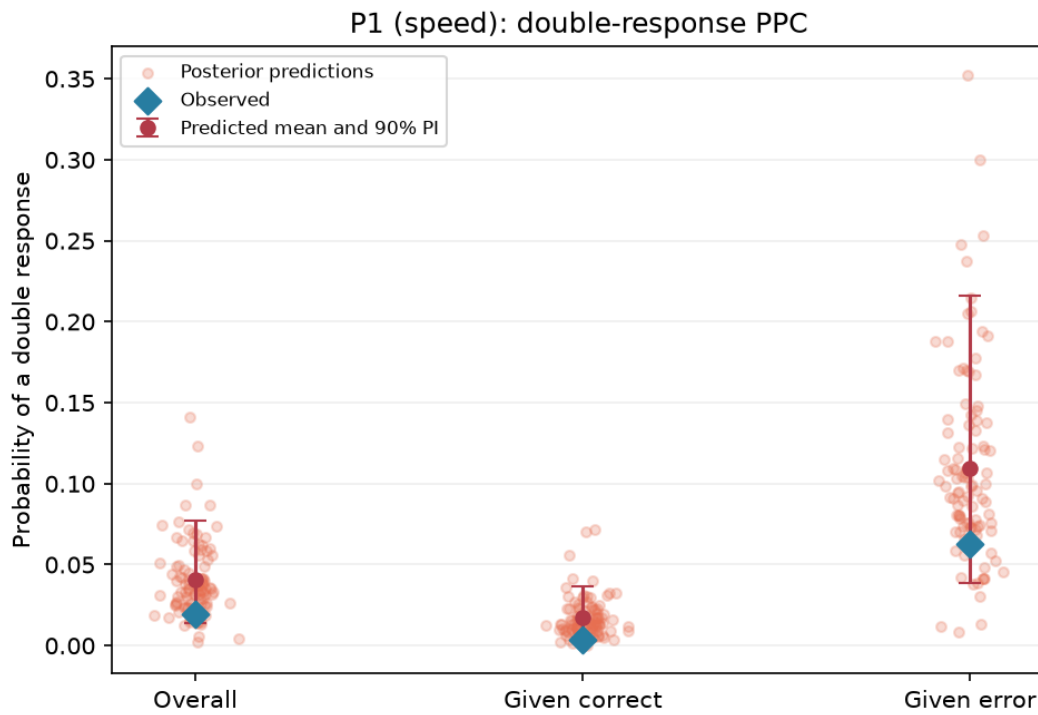


Figure 20: Posterior predictive double-response rates for participant 1, including overall and conditional rates.

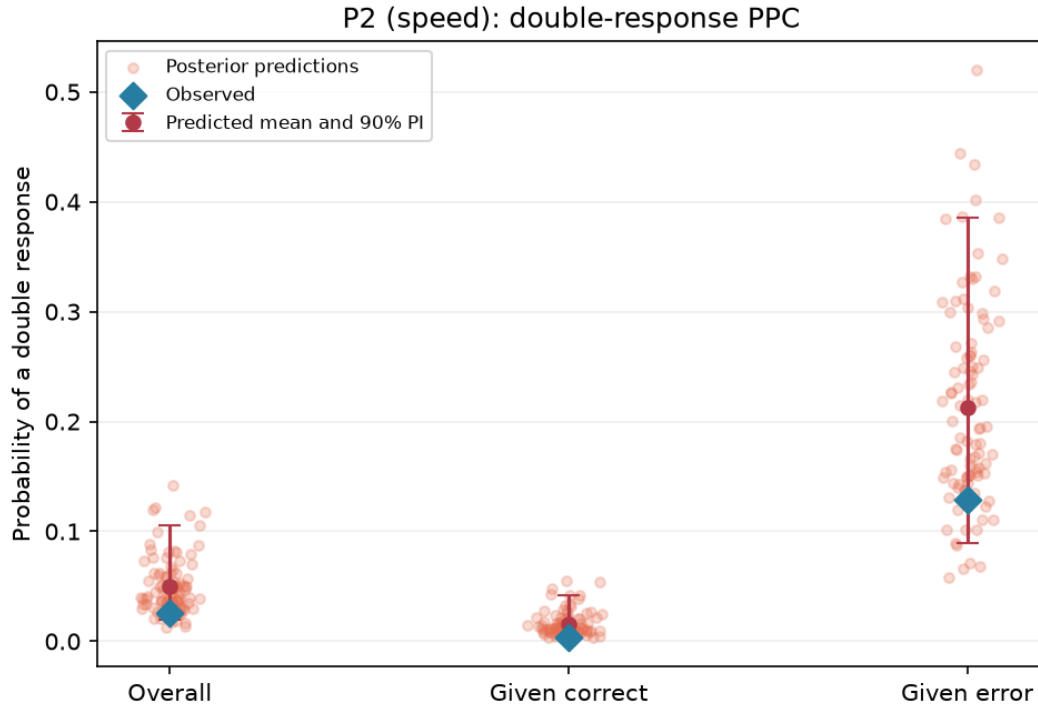


Figure 21: Posterior predictive double-response rates for participant 2, including overall and conditional rates.

7.3 Double-response-time predictions

For P1, 189 double responses were observed, compared with approximately 403 predicted per posterior draw. For P2, 255 were observed, compared with approximately 498 predicted per draw. For P3, 3 were observed, compared with approximately 7 predicted per draw. For P4, 7 were observed, compared with approximately 12 predicted per draw. Because so few double responses occurred for these two participants, their observed double-response-time distributions are too sparse for a stable comparison, even though the model's own predictions remain reasonably populated.

For P1 and P2, where the comparison is stable, the mismatch has a clear shape: the observed second responses are concentrated early in the 250 ms window, while the model's predicted distribution is comparatively flat and places more probability on longer delays. The model therefore reproduces that a second response tends to follow quickly, but it does not reproduce how quickly.

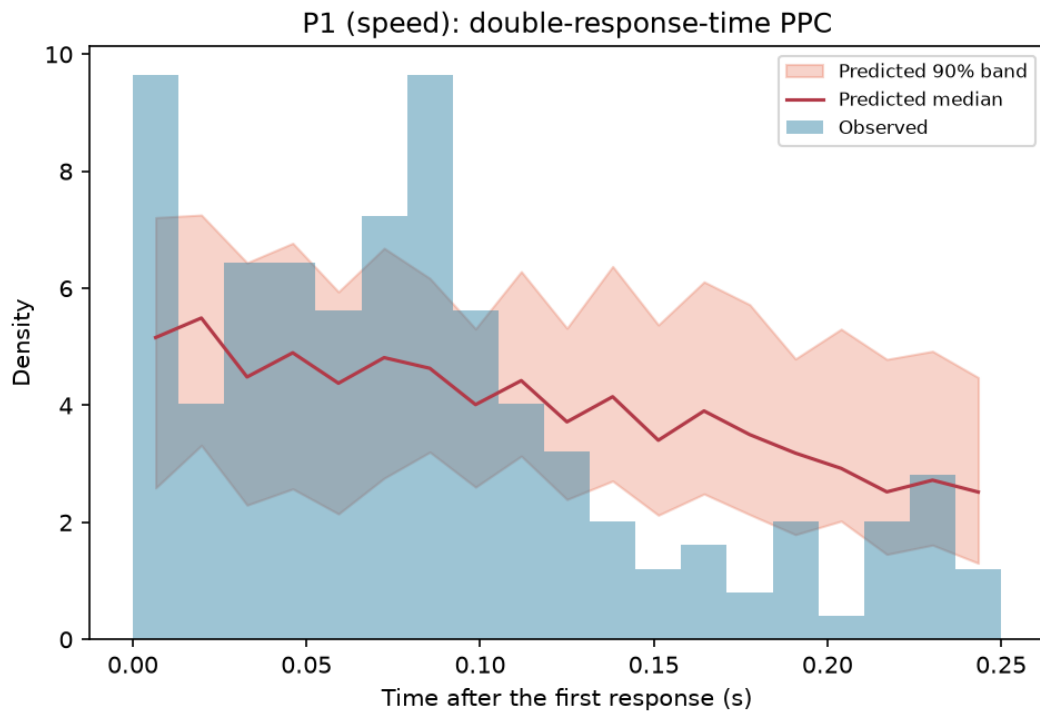


Figure 22: Posterior predictive distribution of the delay between the first and second responses for participant 1.

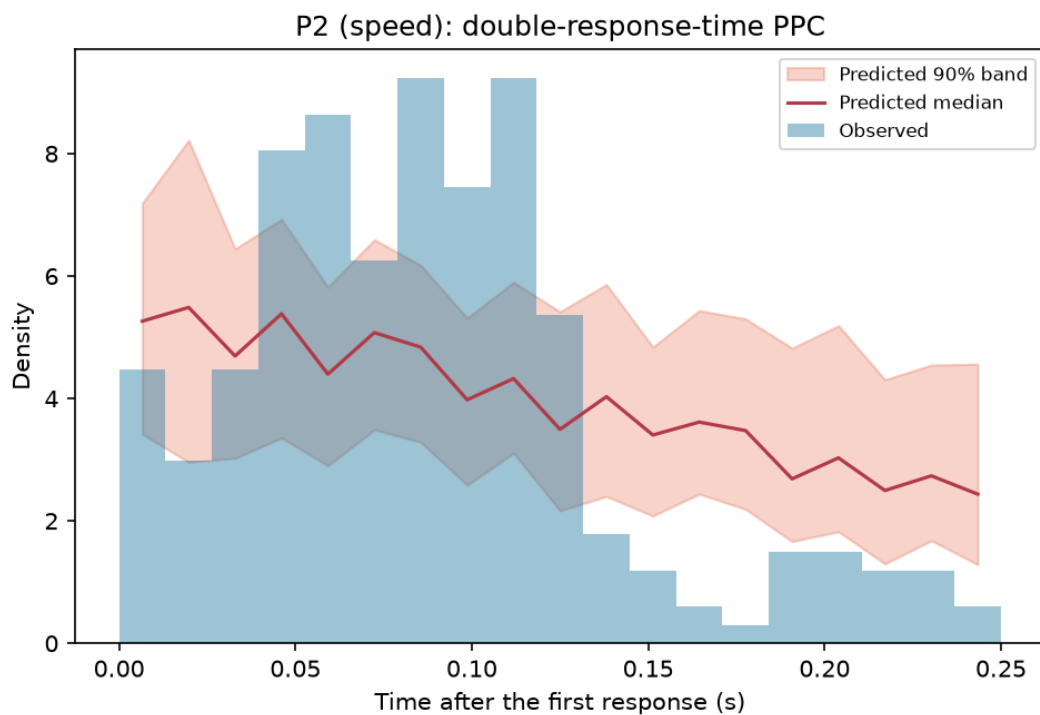


Figure 23: Posterior predictive distribution of the delay between the first and second responses for participant 2.

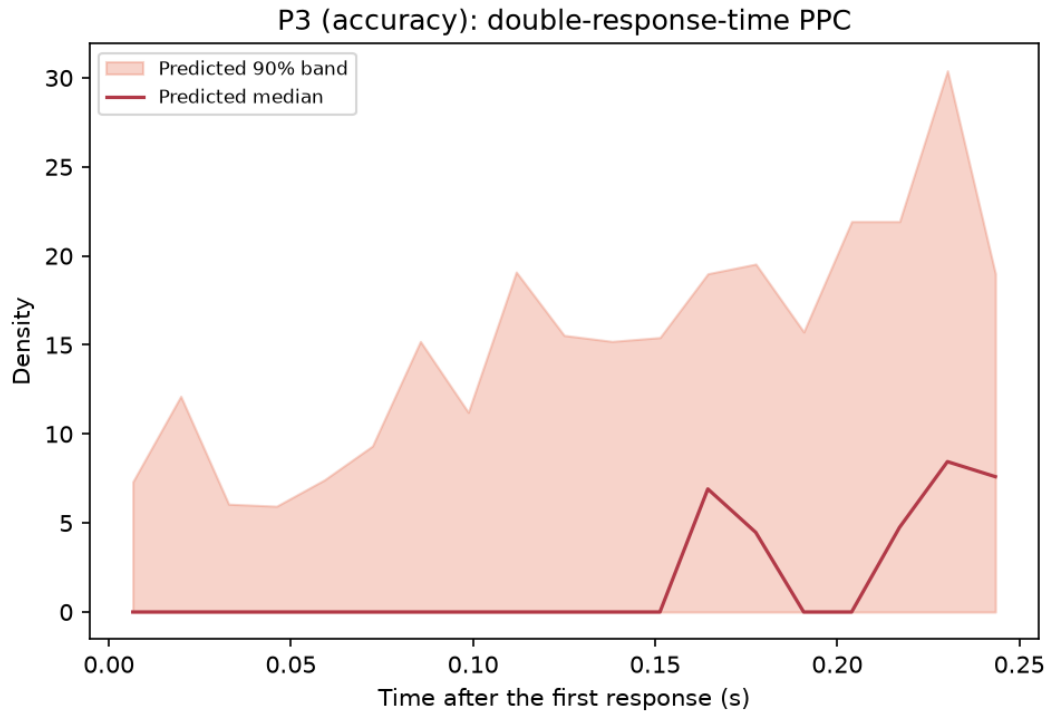


Figure 24: Double-response-time comparison for participant 3. With only 3 observed double responses, an observed distribution could not be meaningfully plotted and is omitted; only the posterior predictive band and median are shown.

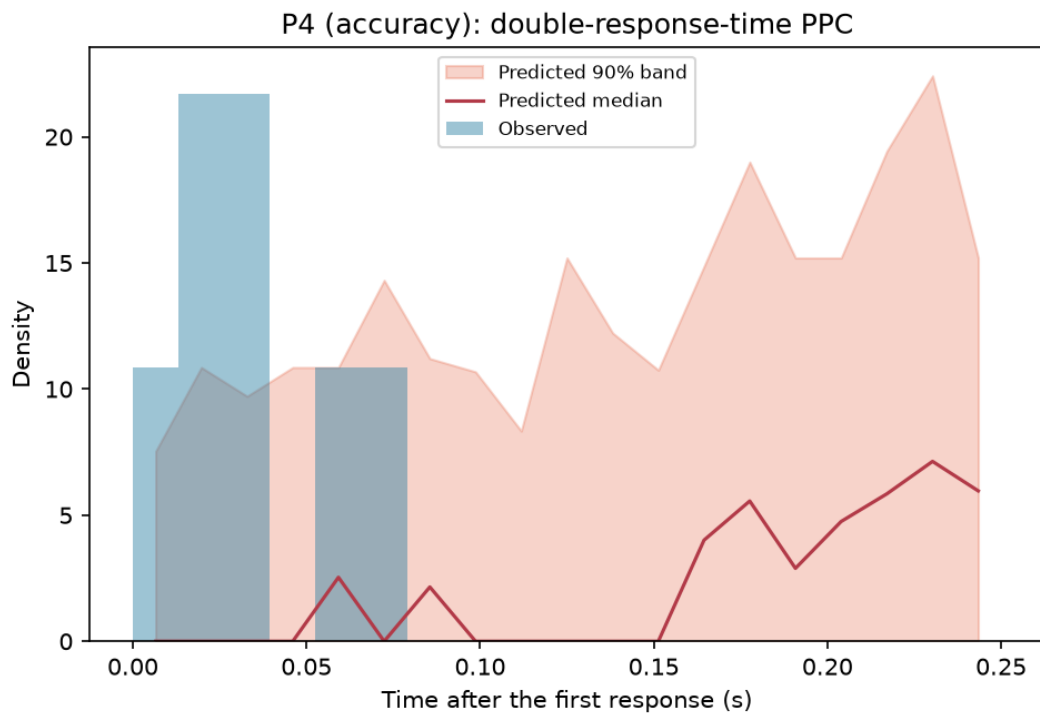


Figure 25: Double-response-time comparison for participant 4.

8 Discussion and limitations

The project shows how double responses can be incorporated into an evidence accumulation model and used as an additional source of information for simulation-based inference. The fitted model reproduces the main empirical differences between speed and accuracy participants and gives close predictions for accuracy and mean response time.

The double-response results are more mixed. The model correctly reproduces the strong asymmetry between double responses after errors and after correct responses, but it predicts too many double responses for the two speed participants. This indicates that the double-response mechanism remains a more difficult part of the data than the standard response-time and accuracy summaries.

The training diagnostics also show why a lower neural-network loss should not be treated as the only convergence criterion. Training to epoch 80 substantially reduced the loss, but the calibration diagnostics did not improve uniformly. We therefore used the epoch-60 checkpoint for empirical inference. In future work, a larger and more targeted simulation budget could be used, particularly around parameter combinations that are difficult to recover. The double-response mechanism could also be examined with alternative definitions or additional model structures considered by Evans et al. [1].

Another limitation is the small number of participants. The empirical analysis concerns four participants, so the results should be interpreted as participant-level model fits rather than population-level conclusions. In addition, P3 and P4 contain very few double responses, limiting the information available for estimating and validating the double-response mechanism in those participants.

9 Conclusion

We implemented a lateral-inhibition racing diffusion model with double responding in BayesFlow and used simulation-based inference to estimate its five parameters from empirical decision data. The analysis included prior predictive checks, extended online training, parameter recovery, SBC, posterior contraction and z-score diagnostics, and posterior predictive checks.

The main empirical behaviour was captured well: the model reproduced accuracy and response times for all four participants and reproduced the strong tendency for double responses to follow errors rather than correct responses. At the same time, the model overpredicted the overall frequency of double responses, particularly in the speed condition. This makes double responding a useful diagnostic: it exposes model mismatch that would be less visible from accuracy and response time alone.

Overall, the project demonstrates the complete SBI workflow from an explicit simulator and priors to amortized posterior inference and predictive validation on real data.

10 Reflection on Our Learning

The project helped us understand that SBI is not simply a replacement for a conventional fitting routine. The central challenge is to build a simulator whose parameters, data-generating process, and preprocessing match the empirical problem closely enough that the neural estimator learns a meaningful inverse mapping.

A particularly important lesson was that a low training loss is not sufficient evidence that an SBI model is ready for empirical inference. Recovery, SBC, posterior contraction, and posterior predictive checks answer different questions. We also learned that preprocessing choices must be mirrored between simulated and empirical data. In this project, the 250 ms response-time exclusion and the explicit double-response indicator were therefore applied consistently.

Finally, the project showed us why uncertainty should be part of the final interpretation. Posterior means alone can hide weakly identified parameters, whereas credible intervals and predictive intervals make it possible to distinguish a precise inference from an uncertain one.

References

- [1] Evans, N. J., Dutilh, G., Wagenmakers, E.-J., & van der Maas, H. L. (2020). Double responding: A new constraint for models of speeded decision making. *Cognitive Psychology*, 121, 101292. <https://doi.org/10.1016/j.cogpsych.2020.101292>.
- [2] Dutilh, G., Vandekerckhove, J., Tuerlinckx, F., & Wagenmakers, E.-J. (2009). A diffusion model decomposition of the practice effect. *Psychonomic Bulletin & Review*, 16, 1026–1036. <https://doi.org/10.3758/16.6.1026>.
- [3] BayesFlow Team. BayesFlow (version 2.0.12) [Computer software]. <https://github.com/bayesflow-org/bayesflow>.